MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCIAM101 Automotive Systems for E- Vehicles

Duration: 2 hours **Maximum Marks:** 35

Practical Examination – Model Question Paper

Instructions:

- Demonstrate the given task. (Any one of the tasks listed below)
- Viva session (Minimum 5 questions from the Practical Module)
- Lab Report

Evaluation Criteria:

- Viva: 7 marks
- Lab Report: 8 marks
- Demonstration: 20 marks

Tasks: (Choose any one)

- ❖ Develop a comparator circuit using op-amp 741 (the reference voltage is 2V)
- ❖ Develop a zero-crossing detector using op-amp 741
- ❖ Develop an inverting amplifier using op-amp 741 to provide a gain 10
- ❖ Develop a non-inverting amplifier using op-amp 741 to provide a gain 11
- ❖ Develop a circuit which provides a DC output of 5V using W10 IC and Capacitor
- ❖ Develop a circuit to amplify the signal from LDR using op-amp 741